



DEPARTAMENTO DE
MATEMÁTICA

The Principles of Deepn Learning Theory

Capítulo 4: RG-flow de pré-ativações

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Introdução

4.1 - 1ª Camada: A boa e velha Gaussiana

“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

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Dataset com inputs n_0 -dimensionais e com N_D amostras:

$$D = \{x_{i,\alpha}\}_{i=1,\dots,n_0;\alpha=1,\dots,N_D}$$

Pré-ativações da 1ª camada são dadas por:

$$z_{i,\alpha}^{(1)} \equiv z_i^{(1)}(x_\alpha) = b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha}, \quad \text{para } i = 1, \dots, n_1.$$

Na inicialização, estabelecemos $b^{(1)}$, $W^{(1)}$ independentemente distribuídas e tais que:

$$\mathbb{E}[b_i^{(1)}] = \mathbb{E}[W_{i_1 j_1}^{(1)}] = 0$$

e

$$\mathbb{E}[b_i^{(1)} b_j^{(1)}] = \delta_{ij} C_b^{(1)}$$

$$\mathbb{E}[W_{i_1 j_1}^{(1)} W_{i_2 j_2}^{(1)}] = \delta_{i_1 i_2} \delta_{j_1 j_2} \frac{C_W^{(1)}}{n_0}$$

Estamos interessados em:

$$p(z^{(1)}|D) = p(z^{(1)}(x_1), \dots, z^{(1)}(x_{N_D}))$$

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Vamos fazer a demonstração via 2 caminhos:

- ▶ combinatório, via contrações de Wick e correlatores;
- ▶ algébrico, via transformada de Hubbard-Stratonovich.

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Vemos primeiramente que:

$$\begin{aligned}\mathbb{E}[z_{i,\alpha}^{(1)}] &= \mathbb{E}[b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha_1}] \\ &= \mathbb{E}[b_i^{(1)}] + \sum_{j=1}^{n_0} \mathbb{E}[W_{ij}] x_{j,\alpha} = 0\end{aligned}$$

dados que as distribuições têm média 0.

É “fácil ver que” todos os correlatores de ordem ímpar de $p(z^{(1)}|D)$ são zerados, porque sempre sobra um $b^{(1)}$ ou $W^{(1)}$ em cada termo que zera a conta.

Para um correlator de dois pontos:

$$\mathbb{E} \left[z_{i_1; \alpha_1}^{(1)} z_{i_2; \alpha_2}^{(1)} \right] = \mathbb{E} \left[\left(b_{i_1}^{(1)} + \sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(b_{i_2}^{(1)} + \sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right]$$

Para um correlator de dois pontos:

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 \mathbb{E} \left[z_{i_1; \alpha_1}^{(1)} z_{i_2; \alpha_2}^{(1)} \right] &= \mathbb{E} \left[\left(b_{i_1}^{(1)} + \sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(b_{i_2}^{(1)} + \sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right]
 \end{aligned}$$

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 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \delta_{i_1 i_2} C_b^{(1)} + \sum_{j_1, j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j, \alpha_1} x_{j, \alpha_2}
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 &= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1, j_1}] x_{j_1, \alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2, j_2}] x_{j_2, \alpha_2} \\
 &\quad + \mathbb{E} \left[\left(\sum_{j_1=1}^{n_0} W_{i_1 j_1}^{(1)} x_{j_1; \alpha_1} \right) \left(\sum_{j_2=1}^{n_0} W_{i_2 j_2}^{(1)} x_{j_2; \alpha_2} \right) \right] \\
 &= \delta_{i_1 i_2} C_b^{(1)} + \sum_{j_1, j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j, \alpha_1} x_{j, \alpha_2} \\
 &= \delta_{i_1 i_2} \left(C_b^{(1)} + C_W^{(1)} \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j; \alpha_1} x_{j; \alpha_2} \right)
 \end{aligned}$$

Definimos a seguinte expressão como a **métrica de primeira camada**:

$$G_{\alpha_1 \alpha_2}^{(1)} := C_b^{(1)} + C_W^{(1)} \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j; \alpha_1} x_{j; \alpha_2}$$

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Note que é uma função de $x_{\alpha_1}, x_{\alpha_2}$. Assim, nossa igualdade acima se traduz como:

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Note que é uma função de $x_{\alpha_1}, x_{\alpha_2}$. Assim, nossa igualdade acima se traduz como:

$$\mathbb{E} \left[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} \right] = \delta_{i_1 i_2} G_{\alpha_1\alpha_2}^{(1)}$$

Seguindo a mesma ideia:

$$\begin{aligned}\mathbb{E}[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)}] &= \delta_{i_1 i_2} \delta_{i_3 i_4} G_{\alpha_1 \alpha_2}^{(1)} G_{\alpha_3 \alpha_4}^{(1)} + \delta_{i_1 i_3} \delta_{i_2 i_4} G_{\alpha_1 \alpha_3}^{(1)} G_{\alpha_2 \alpha_4}^{(1)} \\ &\quad + \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(1)} G_{\alpha_2 \alpha_3}^{(1)}\end{aligned}$$

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 &\quad + \delta_{i_1 i_4} \delta_{i_2 i_3} G_{\alpha_1 \alpha_4}^{(1)} G_{\alpha_2 \alpha_3}^{(1)} \\
 &= \mathbb{E} \left[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} \right] \mathbb{E} \left[z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)} \right] + \mathbb{E} \left[z_{i_1;\alpha_1}^{(1)} z_{i_3;\alpha_3}^{(1)} \right] \mathbb{E} \left[z_{i_2;\alpha_2}^{(1)} z_{i_4;\alpha_4}^{(1)} \right] \\
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 \end{aligned}$$

Lembrando de discussão anterior, isso significa que

$$\mathbb{E}[z_{i_1;\alpha_1}^{(1)} z_{i_2;\alpha_2}^{(1)} z_{i_3;\alpha_3}^{(1)} z_{i_4;\alpha_4}^{(1)}]_{\text{connected}} = 0$$

de modo que a distribuição é Gaussiana.

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De forma similar, podemos verificar que todos os correlatores conexos de ordem mais alta zeram, e portanto, podem ser gerados por uma Gaussiana.

Lembrando, uma distribuição Gaussiana multivariada z de média s e matriz de covariância K pode ser escrita como

$$\begin{aligned} p(z) &= \frac{1}{\sqrt{|2\pi K|}} \exp \left[-\frac{1}{2} \sum_{\mu, \nu=1}^N (z-s)_{\mu} K^{\mu\nu} (z-s)_{\nu} \right] \\ &= \frac{1}{\sqrt{|2\pi K|}} \exp \left[-\frac{1}{2} (z-s)^T K^{-1} (z-s) \right] \end{aligned}$$

onde $K^{\mu\nu} = (K^{-1})_{\mu\nu}$.

Para escrever a distribuição de primeira camada, precisamos da inversa de $\delta_{i_1 i_2} G_{(1)}^{\alpha_1 \alpha_2}$, para cada par de índices i_1, i_2 , que satisfaça

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$$\sum_{j=1}^{n_1} \sum_{\beta \in D} (\delta_{i_1 j} G_{(1)}^{\alpha_1 \beta}) (\delta_{i_2 j} G_{\beta \alpha_2}^{(1)}) = \delta_{i_1 i_2} \delta_{\alpha_2}^{\alpha_1}$$

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$G_{(1)}^{\alpha\beta}$ denota as entradas da matriz $(G^{(1)})_{\alpha\beta}^{-1} \in \mathbb{R}^{N_D}$, com as métricas $G_{\alpha\beta}^{(1)}$ para duas entradas x_α, x_β , satisfazendo

$$\sum_{\beta \in D} G_{(1)}^{\alpha_1 \beta} G_{\beta \alpha_2}^{(1)} = \delta_{\alpha_2}^{\alpha_1}$$

Podemos, então, escrever a distribuição $p(z^{(1)}|D)$ na forma:

$$p(z^{(1)}|D) = \frac{1}{Z} e^{-S(z^{(1)})}$$

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$$S(z^{(1)}) = \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z_{i;\alpha_1}^{(1)} G_{(1)}^{\alpha_1 \alpha_2} z_{i;\alpha_2}^{(1)}$$

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$$\left(= \sum_{i,j=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z_{i;\alpha_1}^{(1)} \delta_{ij} G_{(1)}^{\alpha_1 \alpha_2} z_{j;\alpha_2}^{(1)} \right)$$

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e Z é a *função de partição*:

$$Z = \int \left[\prod_{i,\alpha} dz_{i,\alpha}^{(1)} \right] e^{-S(z^{(1)})} = |2\pi G^{(1)}|^{\frac{n_1}{2}}$$

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Ação Gaussiana em ação

Como visto no capítulo anterior, podemos expressar a distribuição na forma

$$p(z \mid D) = \int \left[\prod_j db_j p(b_j) \right] \left[\prod_{j,k} dW_{jk} p(W_{jk}) \right] \prod_{j,\alpha} \delta \left(z_{j;\alpha} - b_i - \sum_j W_{jk} x_{j;\alpha} \right)$$

suprimindo por hora os sobrescritos (1).

Por hipótese:

$$p(b_j) = \frac{1}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right)$$

$$p(W_{jk}) = \frac{1}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right)$$

Vamos usar a **transformação de Hubbard-Stratonovich**. I.e., a representação integral do delta de Dirac:

$$\delta(z - a) = \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)}$$

Obtemos:

$$p(z \mid D) = \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right) \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right) \right] \left[\prod_{j,\alpha} \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)} \right]$$

Obtemos:

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \exp\left(-\frac{b_j^2}{2C_b}\right) \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \exp\left(-\frac{n_0 W_{jk}^2}{2C_W}\right) \right] \left[\prod_{j,\alpha} \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)} \right] \\
 &= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{i,j} \frac{dW_{ij}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{i,\alpha} \frac{d\Lambda_i^\alpha}{2\pi} \right] \\
 &\quad \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{j,k} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right]
 \end{aligned}$$

Completando quadrados:

$$\begin{aligned}
 & \sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{i,j} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{k;\alpha} \right) \\
 &= -\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 - \frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i \frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \\
 & \quad - \frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha}
 \end{aligned}$$

Voltando à exponencial,

$$\int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\ \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right]$$

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 & \times \exp \left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha} \right) \right] \\
 & = \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
 & \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
 & \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
 \end{aligned}$$

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&= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
&\exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
\end{aligned}$$

$$\begin{aligned}
&= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \\
&\exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
&= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}} \right] \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 \right] \\
&\times \int \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \right] \exp \left[-\frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i\frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 \right] \\
&\int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
\end{aligned}$$

Olhando bem, são integrais Gaussianas deslocadas por uma média:

$$\begin{aligned} \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \right] \exp \left[-\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_{\alpha} \Lambda_j^{\alpha} \right)^2 \right] \\ = \prod_j \int \left[\frac{db_j}{\sqrt{2\pi C_b}} \exp \left(-\frac{(b_j - \mu_b)^2}{2C_b} \right) \right] \end{aligned}$$

com a média $\mu_b = -iC_b \sum_{\alpha} \Lambda_j^{\alpha}$. Assim, cada integral dessa é igual a 1, e some da representação.

Olhando bem, são integrais Gaussianas deslocadas por uma média:

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com a média $\mu_b = -iC_b \sum_{\alpha} \Lambda_j^{\alpha}$. Assim, cada integral dessa é igual a 1, e some da representação.

De forma análoga, a integral envolvendo W_{jk} vira 1.

Então...

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_{\alpha} \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_{\alpha} \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp \left[-\frac{C_W}{2n_0} \sum_{j,k} \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} x_{k;\alpha_1} x_{k;\alpha_2} - \frac{C_b}{2} \sum_j \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[-\frac{1}{2} \sum_{j,\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} \left(C_b + C_W \frac{1}{n_0} \sum_k x_{k;\alpha_1} x_{k;\alpha_2} \right) + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right] \\
 &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[-\frac{1}{2} \sum_{j,\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$-\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j;\alpha}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned} -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j;\alpha} \\ = -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned}
 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j;\alpha} \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda \right]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned}
 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha} \\
 & \quad = -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \\
 & \quad = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z) \right]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned}
 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha} \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z) \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G (\Lambda - i G^{-1} z) - i (G^{-1} z)^T G (\Lambda - G^{-1} z) + z^T G^{-1} z \right]
 \end{aligned}$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$\begin{aligned}
 & -\frac{1}{2} \sum_{j, \alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j, \alpha} \Lambda_j^\alpha z_{j; \alpha} \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z) \right] \\
 & = -\frac{1}{2} \left[\Lambda^T G (\Lambda - i G^{-1} z) - i (G^{-1} z)^T G (\Lambda - G^{-1} z) + z^T G^{-1} z \right] \\
 & = -\frac{1}{2} \left[(\Lambda - i G^{-1} z)^T G (\Lambda - i G^{-1} z) + z^T G^{-1} z \right].
 \end{aligned}$$

Ou seja,

$$-\frac{1}{2}\Lambda^T G \Lambda + i\Lambda^T z = -\frac{1}{2} [(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) + z^T G^{-1}z].$$

Em notação de somatório, para ficar igual ao livro

$$\begin{aligned} & -\frac{1}{2} [(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) + z^T G^{-1}z] \\ &= -\frac{1}{2} \sum_{j, \alpha_1, \alpha_2} \left[G_{\alpha_1 \alpha_2}^{(1)} \left(\Lambda_{\alpha_1 j} - i \sum_{\beta_1} G_{(1)}^{\alpha_1 \beta_1} z_{j; \beta_1}^{(1)} \right) \left(\Lambda_{\alpha_2 j} - i \sum_{\beta_2} G_{(1)}^{\alpha_2 \beta_2} z_{j; \beta_2}^{(1)} \right) + G_{(1)}^{\alpha_1 \alpha_2} z_{j; \alpha_1}^{(1)} z_{j; \alpha_2}^{(1)} \right] \end{aligned}$$

Assim,

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) \right] \exp \left[z^T G^{-1}z \right] \\
 &= |2\pi G^{-1}|^{\frac{n_1}{2}} \exp \left[z^T G^{-1}z \right] \\
 &= \frac{1}{|2\pi G^{(1)}|^{\frac{n_1}{2}}} \exp \left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)} \right]
 \end{aligned}$$

voltando ao final o sobrescrito e a notação de somatório.

Assim,

$$\begin{aligned}
 p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp \left[(\Lambda - iG^{-1}z)^T G (\Lambda - iG^{-1}z) \right] \exp \left[z^T G^{-1}z \right] \\
 &= |2\pi G^{-1}|^{\frac{n_1}{2}} \exp \left[z^T G^{-1}z \right] \\
 &= \frac{1}{|2\pi G^{(1)}|^{\frac{n_1}{2}}} \exp \left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)} \right]
 \end{aligned}$$

voltando ao final o sobrescrito e a notação de somatório.

Ou seja, o mesmo termo que encontramos na seção “Wick this way”.

Introdução

4.1 - 1ª Camada: A boa e velha Gaussiana

“Wick this way”, derivação combinatória

”Hubbard-Stratonovich this way”: derivação algébrica via ação

Ação Gaussiana em ação

$$= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \right]$$

$$\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right]$$

$$= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j;\beta_1} z_{j;\beta_2} \right) \sigma(z_{i_1;\alpha_1}) \sigma(z_{i_1;\alpha_2}) \mathbb{E} \left[\sigma \left(z_{i_1;\alpha_1}^{(1)} \right) \sigma \left(z_{i_1;\alpha_2}^{(1)} \right) \right]$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\}
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\} \\
 &\quad \times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2})
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\} \\
 &\quad \times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \{1\} \times \left[\int \prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2})
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] \\
 &= \int \left[\prod_{i=1}^{n_1} \prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \times \exp \left(-\frac{1}{2} \sum_{j=1}^{n_1} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{j; \beta_1} z_{j; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \left\{ \prod_{i \neq i_1} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left[-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{\beta_1 \beta_2}^{(1)} z_{i; \beta_1} z_{i; \beta_2} \right] \right\} \\
 &\quad \times \int \left[\prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &= \{1\} \times \left[\int \prod_{\alpha \in D} \frac{dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &:= \langle \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \rangle_{G^{(1)}}.
 \end{aligned}$$

Com a expressão do final, reintroduzimos a notação:

$$\langle F(z_{\alpha_1}, \dots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\Pi_{\alpha \in D} dz_{\alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \dots, z_{\alpha_m})$$

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$$\langle F(z_{\alpha_1}, \dots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_{\alpha}}{\sqrt{|2\pi g|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \dots, z_{\alpha_m})$$

e denotamos ainda:

$$\sigma_{\alpha} := \sigma(z_{\alpha})$$

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e denotamos ainda:

$$\sigma_{\alpha} := \sigma(z_{\alpha})$$

Assim, podemos escrever

$$\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \right] = \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \rangle_{G^{(1)}}$$

É possível ainda generalizar para correladores de mais pontos.
Para todas as saídas do mesmo neurônio i_1 :

$$\mathbb{E} \left[\sigma(z_{i_1; \alpha_1}^{(1)}) \sigma(z_{i_1; \alpha_2}^{(1)}) \sigma(z_{i_1; \alpha_3}^{(1)}) \sigma(z_{i_1; \alpha_4}^{(1)}) \right] = \langle \sigma_{\alpha_1} \sigma_{\alpha_2} \sigma_{\alpha_3} \sigma_{\alpha_4} \rangle_{G^{(1)}}$$

fazendo as exatas mesmas etapas.

Para $i_1 \neq i_2$

$$\begin{aligned}
 & \mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right] \\
 &= \left\{ \prod_{i \notin \{i_1, i_2\}} \int \left[\prod_{\alpha \in D} \frac{dz_{i; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i; \beta_1} z_{i; \beta_2} \right) \right\} \\
 &\times \int \left[\frac{\prod_{\alpha \in D} dz_{i_1; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_1; \beta_1} z_{i_1; \beta_2} \right) \sigma(z_{i_1; \alpha_1}) \sigma(z_{i_1; \alpha_2}) \\
 &\times \int \left[\frac{\prod_{\alpha \in D} dz_{i_2; \alpha}}{\sqrt{|2\pi G^{(1)}|}} \right] \exp \left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} G_{(1)}^{\beta_1 \beta_2} z_{i_2; \beta_1} z_{i_2; \beta_2} \right) \sigma(z_{i_2; \alpha_3}) \sigma(z_{i_2; \alpha_4}) \\
 &= \left\langle \sigma(z_{\alpha_1}) \sigma(z_{\alpha_2}) \right\rangle_{G^{(1)}} \left\langle \sigma(z_{\alpha_3}) \sigma(z_{\alpha_4}) \right\rangle_{G^{(1)}}
 \end{aligned}$$

I.e.:

$$\mathbb{E} \left[\sigma \left(z_{i_1; \alpha_1}^{(1)} \right) \sigma \left(z_{i_1; \alpha_2}^{(1)} \right) \sigma \left(z_{i_2; \alpha_3}^{(1)} \right) \sigma \left(z_{i_2; \alpha_4}^{(1)} \right) \right] = \left\langle \sigma(z_{\alpha_1}) \sigma(z_{\alpha_2}) \right\rangle_{G(1)} \left\langle \sigma(z_{\alpha_3}) \sigma(z_{\alpha_4}) \right\rangle_{G(1)}$$

Isso ilustra a independência entre as distribuições de neurônios diferentes.

Obrigado!

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Repositório com os materiais da apresentação:
<https://github.com/felipekriffel/Mestrado>